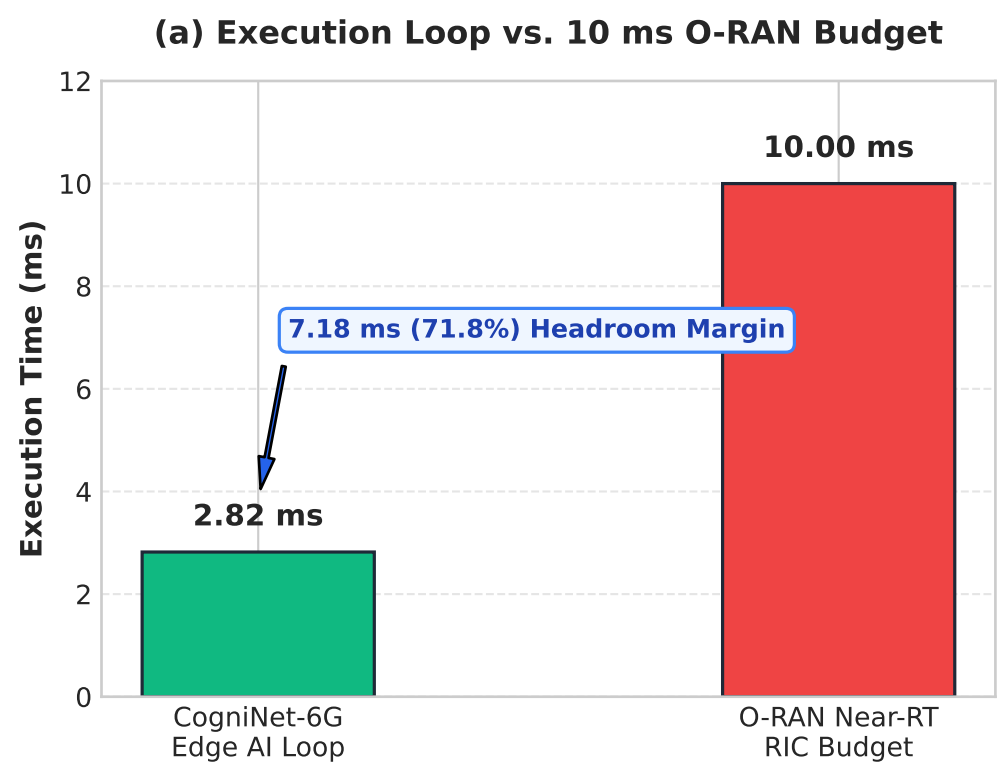
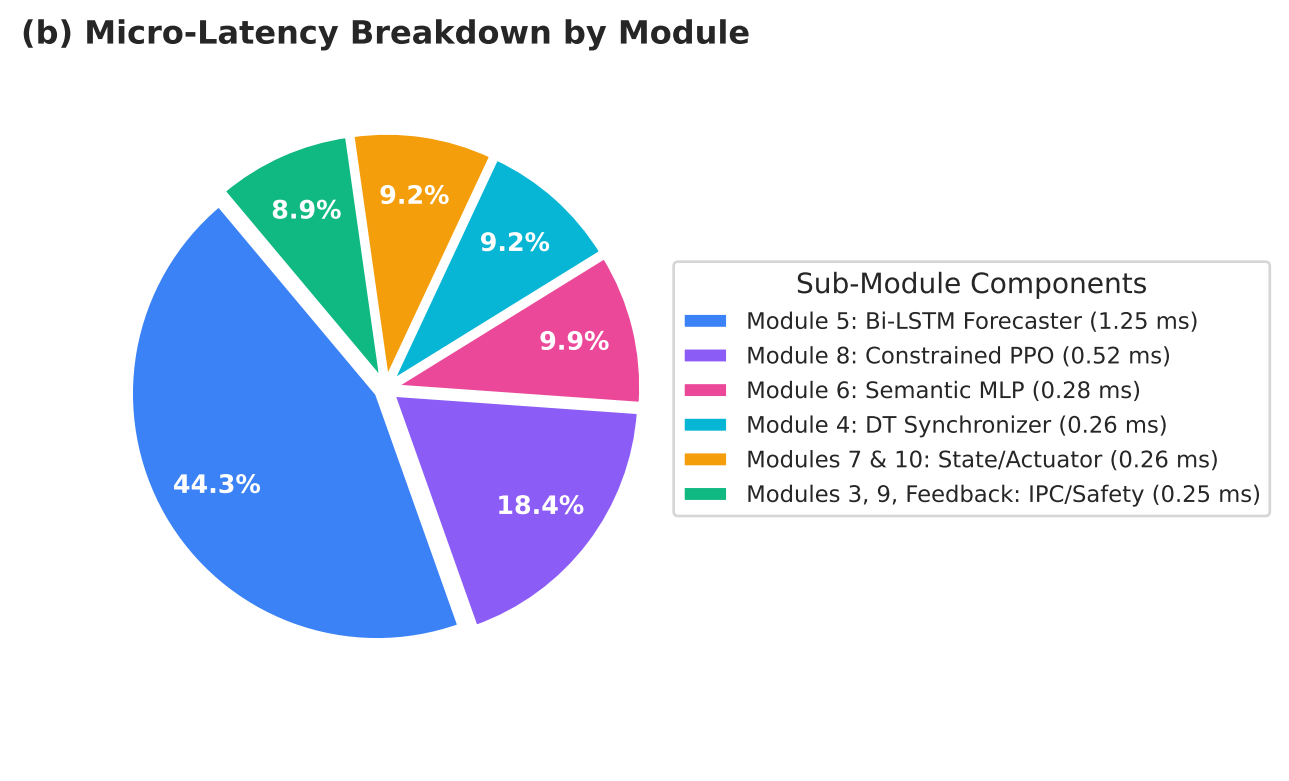

(a) Execution Loop vs. 10 ms O-RAN Budget
Execution Time (ms)
0
2
4
6
8
10
12
10.00 ms
7.18 ms (71.8%) Headroom Margin
2.82 ms
CogniNet-6G
Edge AI Loop
O-RAN Near-RT
RIC Budget


(b) Micro-Latency Breakdown by Module
44.3%
18.4%
9.9%
9.2%
9.2%
8.9%
Sub-Module Components
Module 5: Bi-LSTM Forecaster (1.25 ms)
Module 8: Constrained PPO (0.52 ms)
Module 6: Semantic MLP (0.28 ms)
Module 4: DT Synchronizer (0.26 ms)
Modules 7 & 10: State/Actuator (0.26 ms)
Modules 3, 9, Feedback: IPC/Safety (0.25 ms)